

# Bond Model

Updated for Rocky version 2023 R1

## 1 Usage details

### 1.1 Purpose of module

The *Bond Model* module allows you to create cylindrical shaped bonds between pairs of neighbor particles. The module also allows you to specify the normal and the tangential strength limits of these bonds, which will break when the external loads acting on them exceed these limits so the particles will not be bonded anymore.

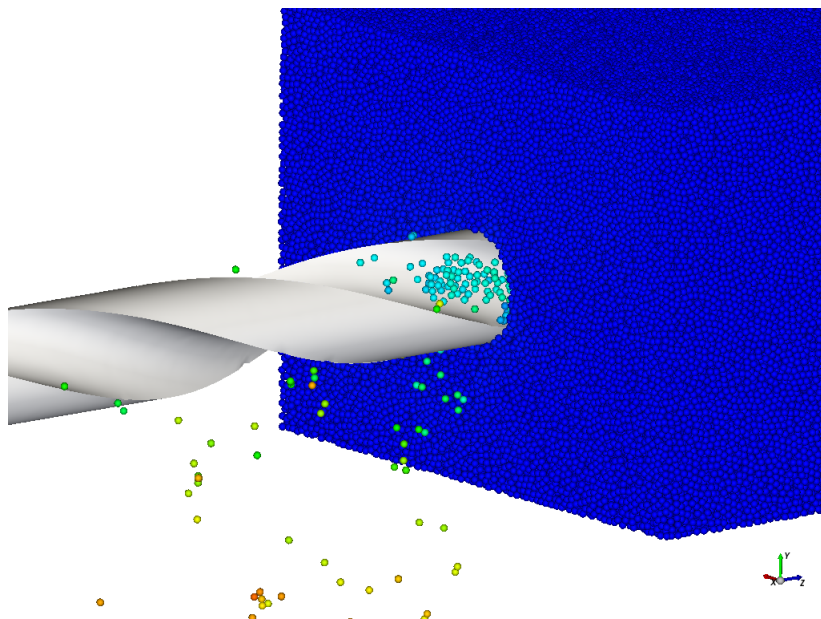


Figure 1: Example application of a drilling process using the *Bond Model* module.

### 1.2 Example application

Figure 1 shows an example of how this module can be applied. The process of drilling a hole in the wall is simulated where neighboring particles are bonded to represent the concrete. As the rotary drill bit forwards against the wall, the bonds strength limits are exceeded and the hole is formed.

### 1.3 Functionality details

This module works by applying particle-to-particle bonds between neighboring particles that have the bond model enabled by you in the Materials Interactions properties and that are also located at a distance lower than the activation distance, which is calculated indirectly according to the distance factor and to the maximum elongation, both defined in the Rocky UI.

The bonds created by the *Bond Model* exert elastic and viscous forces and moments on the neighboring particles as a reaction to deformations caused by their relative motion. Moreover, the bond model includes also the possibility of adding particle-to-boundary bonds to the simulation.<sup>1</sup>

<sup>1</sup> Refer to the [Technical details](#) section for more information.

### 1.4 Limitations and considerations

This module is not compatible with Multiple Element (flexible) particles<sup>2</sup> nor with Coarse Grain Modeling (CGM).<sup>3</sup>

Internally, the module associates every bond in the simulation to an adhesive contact and, as a consequence, Rocky recognizes the module as an adhesive model. Therefore, even if there are no adhesion effects enabled by this module, the iterative method being used takes the place of an adhesion model and will therefore override what you have set for **Adhesive Force** on the **Momentum** tab of the **Physics** entity (Figure 2).

<sup>2</sup> Refer to the *About Adding and Editing Particle Sets* topic in the *Rocky User Manual* for details on flexible particles.

<sup>3</sup> Refer to the *About Physics Parameters* topic in the *Rocky User Manual* for details.

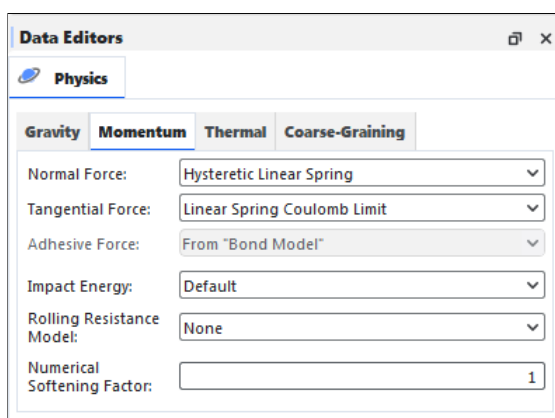


Figure 2: *Physics | Momentum* tab options when the *Bond Model* module is enabled.

Simulations using the *Bond Model* require the contacts data collection to be enabled by the **Collect Contacts Data** checkbox from **Contacts** entity (Figure 3). Moreover, these simulations can be

resumed if you enable the option to save the adhesive contact data. To do that, you must also enable the **Include Adhesive Contacts** checkbox for **Contacts**.

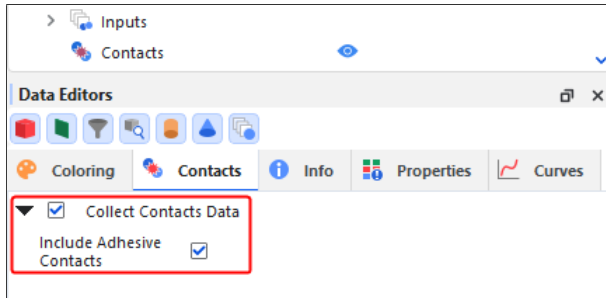


Figure 3: *Contacts* tab options.

Simulations using the *Bond Model* module cannot be saved as project for Restart.

If you are running a simulation with both *ModuleName* and *Particle Freeze Outside Box* enabled, keep in mind that any bond that leaves the region of interest will be lost. Thus, particles that were bonded will no longer have bonds if they reenter the region of interest.

## 1.5 Module options

This module has one parameter that you can set, which is defined below and shown in Figure 4.

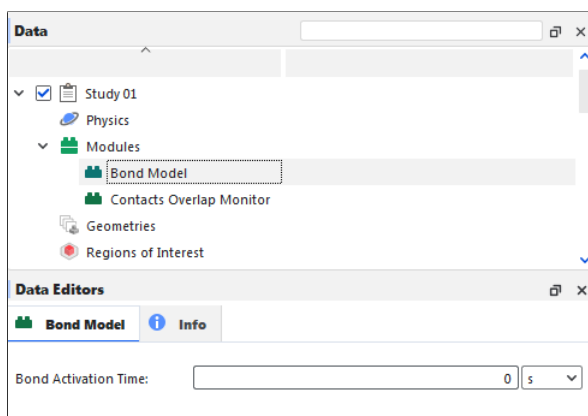


Figure 4: *Bond Model* module options.

- **Bond Activation Time:** Defines the specific time when the bonds activation happens.  
*Range:* [Positive values.]

### 1.5.1 Additional options enabled in the UI

Besides the parameter you define in the module itself, there are additional parameters you must set for each materials interaction. See the section below for details.

#### 1.5.1.1 Additional Material Interactions options

There are seven additional parameters that you must set in the Materials Interactions entity, as shown in Figure 5.

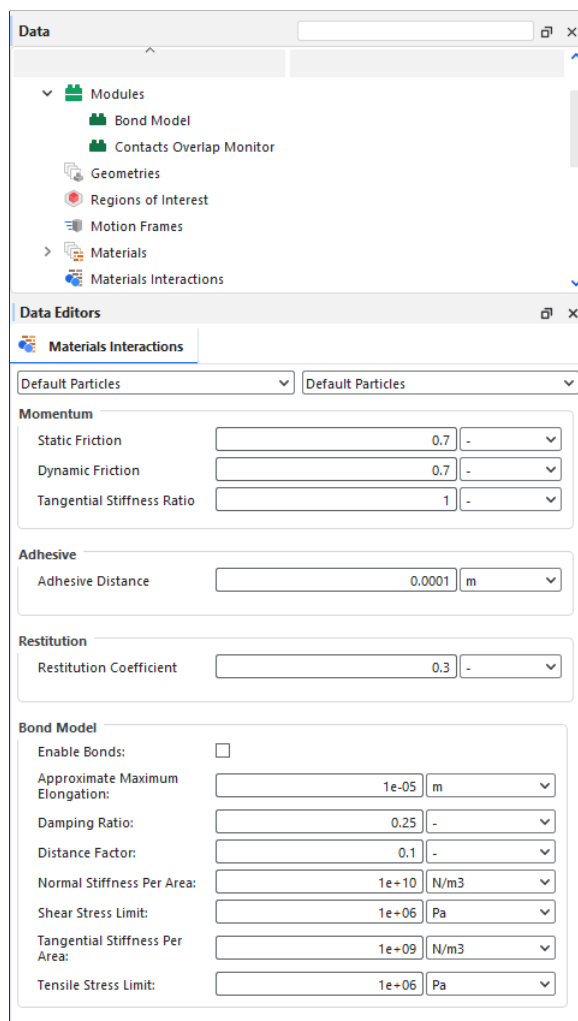


Figure 5: *Bond Model* module materials interactions options.

These parameters are explained below:

- **Enable Bonds:** This enables the bonds for the particles associated with the selected materials.

*Range:* [Turns on or off]

- **Approximate Maximum Elongation:** This defines the maximum elongation for the bonds activation distance calculation.<sup>4</sup>  
*Range:* [Positive values]
- **Damping Ratio:** This defines the damping ratio attributed to the bond.<sup>4</sup>  
*Range:* [Positive values]
- **Distance Factor:** This defines the factor for the bonds activation distance calculation.<sup>4</sup>  
*Range:* [Positive values]
- **Normal Stiffness Per Area:** This defines the normal stiffness per area, which is used to calculate the normal component of the linear elastic force applied on the particles in each bonded pair.<sup>4</sup>  
*Range:* [Positive values]
- **Shear Stress Limit:** This defines the shear stress limit of the bonds.<sup>4</sup>  
*Range:* [Positive values]
- **Tangential Stiffness Per Area:** This defines the tangential stiffness per area, which is used to calculate the tangential component of the linear elastic force applied on the particles in each bonded pair.<sup>4</sup>  
*Range:* [Positive values]
- **Tensile Stress Limit:** This defines the tensile stress limit of the bonds.<sup>4</sup>  
*Range:* [Positive values]

<sup>4</sup> Refer to the [Technical details](#) section for more information.

Since there is no adhesion effects enabled by this module, the parameter **Adhesive Distance** will not be used in any calculations.

## 1.6 Post-processing abilities

After processing your simulation, you can analyze the results using the same contact-related properties that are available by default in Rocky.<sup>5</sup>

In addition, seven extra contact properties are available as shown in Figure 6 and explained below.

<sup>5</sup> Refer to the *About Properties* topic in the *Rocky User Manual* for definitions.

- **Bond Force : X:** is the bond force component in the X direction.

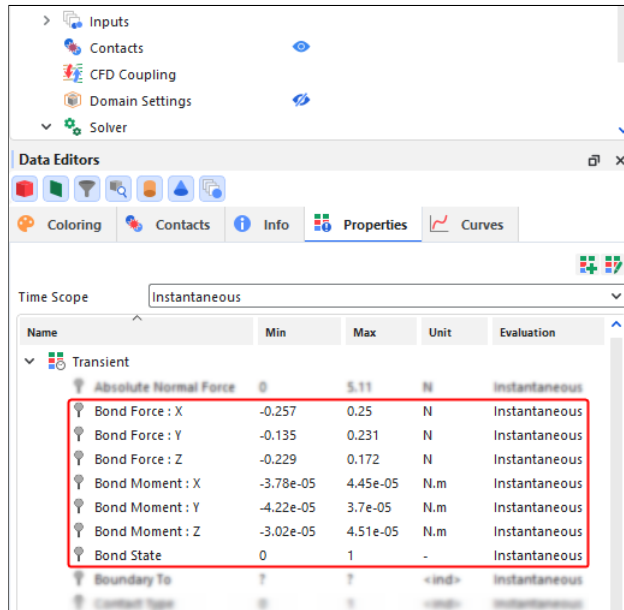


Figure 6: Additional Contact Properties created by the *Bond Model* module.

- **Bond Force : Y:** is the bond force component in the Y direction.
- **Bond Force : Z:** is the bond force component in the Z direction.
- **Bond Moment : X:** is the bond moment component in the X direction.
- **Bond Moment : Y:** is the bond moment component in the Y direction.
- **Bond Moment : Z:** is the bond moment component in the Z direction.
- **Bond State:** Is a criterion that classifies whether the bond is active, inactive or broken. When equals to 0, the bond is inactive; when equals to 1, the bond is active; and when equals to 2, the bond is broken.

Moreover, an additional property is available for the particles as shown in Figure 7 and explained below.

- **Particle Broken Bonds:** The number of broken bonds for the particle.

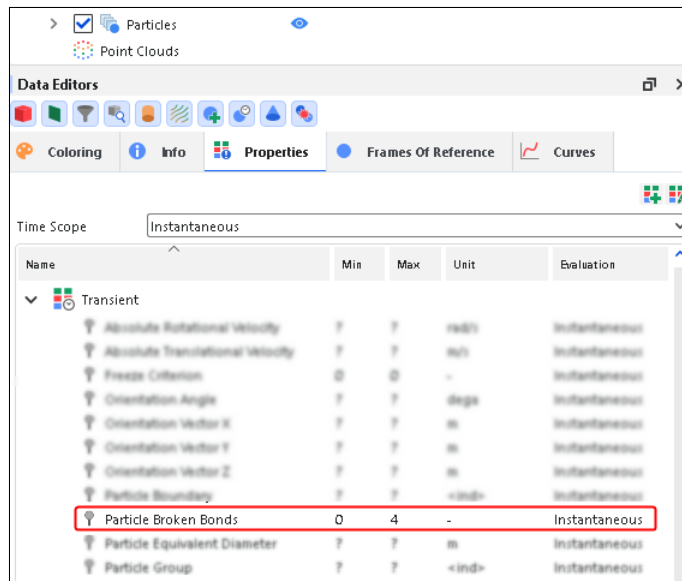


Figure 7: Additional Particle Property created by the *Bond Model* module.

## 1.7 Setting up and using the module

1. Ensure that the module is enabled. (From the **Data** panel, select **Modules** and then from the **Data Editors** panel, ensure the **Bond Model** checkbox is enabled.)
2. From the **Data** panel, under **Modules**, select **Bond Model**.
3. From the **Data Editors** panel, define the **Bond Activation Time**.<sup>6</sup>
4. Continue setting up the project as you normally would.<sup>7</sup>
5. Prior to processing your simulation, ensure that you do the following for each material that will be assigned to particles that you want to have bonds in your project:
  - a. From the **Data** panel, under **Materials Interactions**, select the materials that you assigned to the particles that will have bonds, and then from the **Data Editors** panel do the following:
    - i. From the **Materials Interactions** tab, enable the **Enable Bonds** option.
    - ii. From the **Materials Interactions** tab, enter the values you want for the bond model parameters.<sup>8</sup>
6. Process your simulation as you normally would.<sup>7</sup>
7. When you are ready to analyze your simulation results, you may choose to make use of any contact-related property and may also make use of the additional module-related properties.<sup>9</sup>

<sup>6</sup> Refer to the [Module options](#) section for definitions.

<sup>7</sup> Refer to the *Rocky User Manual* for set up details.

<sup>8</sup> Refer to the [Additional options enabled in the UI](#) section for definitions.

<sup>9</sup> Refer to the [Post-processing abilities](#) section for definitions.

## 2 Technical details

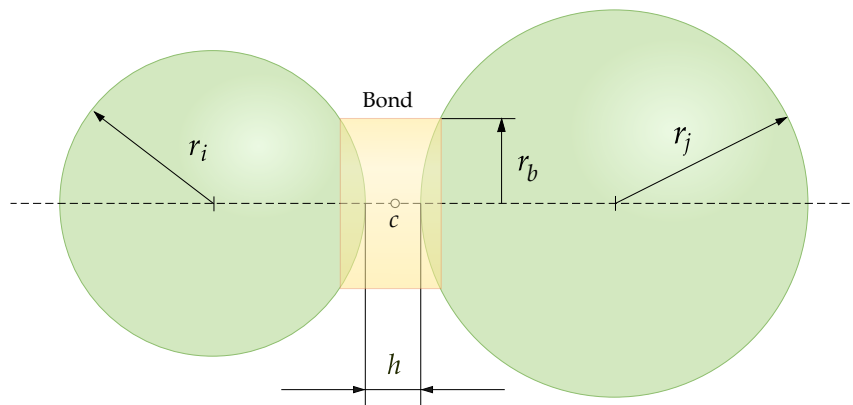
The bond model implemented in this module is similar to the models described in the papers of Gimenez et al.<sup>10</sup> and Potyondy & Cundall.<sup>11</sup> In this model, a bond is a massless entity of cylindrical shape attached to a pair of neighboring particles, that exerts elastic and viscous forces and moments on them as a reaction to deformations caused by their relative motion. If the external load acting on a bond exceeds its specified strength, it will break and its bonding action over the particles will cease.

Bonds are created between pairs of particles satisfying the following two conditions:

- Bonds are explicitly enabled for the *material interaction* associated to the pair of particles by turning on the checkbox **Enable Bonds** in the Rocky UI.
- The closest points of the particles pair surfaces have a distance  $h$  from each other, which is lower than the activation distance given by:

$$h_{ac} = f_{ac} (r_i + r_j) + \delta l \quad (1)$$

where  $r_i$  and  $r_j$  are the radii of the bonded particles, as shown in Figure 8,<sup>12</sup> while  $f_{ac}$  is an input parameter listed as **Distance Factor** in the Rocky UI and  $\delta l$  is another input parameter listed as **Approximate Maximum Elongation**, which is added to compensate the bond tensile deformations during the simulation. The activation of bonds happens only once during a simulation, at a specific time defined by the value of the parameter listed as **Bond Activation Time** in the Rocky UI.<sup>13</sup>



<sup>10</sup> Gimenez, C. S., Finke, B., Nowak, C., Schilde, C., and Kwade, A. (2018). Structural and mechanical characterization of lithium-ion battery electrodes via DEM simulations. *Advanced Powder Technology*, 29(10):2312–2321

<sup>11</sup> Potyondy, D. O. and Cundall, P. A. (2004). A bonded-particle model for rock. *International Journal of Rock Mechanics & Mining Sciences*, 41:1329–1364

<sup>12</sup> For non-spherical particles, the radius is replaced by one half of the sieve size in equation (1) and other equations including it.

<sup>13</sup> Since Rocky calculates forces and solves motion equations at discrete times, the actual activation of the bonds will happen at the discrete time nearest to the specified activation time.

Figure 8: Geometry of the bond between two spherical particles of different sizes.



The radius of the bond between two particles depends only on the radii of those particles, by means of the relationship:

$$r_b = \frac{r_i r_j}{r_i + r_j} \quad (2)$$

The bond model implemented in this module includes also the possibility of adding particle-to-boundary bonds to the simulation. Those bonds share all characteristics described earlier for particle-to-particle bonds, except that the activation distance in this case is given simply by:

$$h_{ac} = 2 f_{ac} r_p + \delta l \quad (3)$$

where  $f_{ac}$  is the corresponding Distance Factor ,  $r_p$  is the radius of the bonded particle and  $\delta l$  is the Approximate Maximum Elongation . Similarly, the radius of the corresponding bond will be directly  $r_b = r_p$  in that case.

At the time of its activation, a bond is in an undeformed state. From that time on, any relative motion of the bonded particles will cause a deformation on the bond, to which it will react exerting forces and moments opposing to that relative motion. The bond deformation can be linear or angular, being the former caused by both the translational and rotational motion of the bonded particles, while the latter is exclusive consequence of their rotational motion. In the module, both types of deformation are calculated incrementally, starting from the time of bond activation,  $t_{ac}$ . Thus, the linear deformation of a bond at any given time  $t$  is given by:

$$\mathbf{s}^b = \sum_{t_{ac}}^t \mathbf{v}_c^{rel} \Delta t \quad (4)$$

where  $\mathbf{v}_c^{rel}$  is the instantaneous relative velocity at the center point of the bond,  $c$ , while  $\Delta t$  is the simulation timestep.<sup>14</sup> The relative velocity  $\mathbf{v}_c^{rel}$  is calculated as:

$$\mathbf{v}_c^{rel} = \mathbf{v}_i - \mathbf{v}_j + \boldsymbol{\omega}_i \times \Delta \mathbf{r}_{i-c} - \boldsymbol{\omega}_j \times \Delta \mathbf{r}_{j-c} \quad (5)$$

where:

- $\mathbf{v}_i$  and  $\mathbf{v}_j$  are the translational velocities of the bonded particles.
- $\boldsymbol{\omega}_i$  and  $\boldsymbol{\omega}_j$  are the rotational velocities of the bonded particles.<sup>15</sup>
- $\Delta \mathbf{r}_{i-c}$  is the vector that joins the centroid of the particle  $i$  to the bond's center point  $c$ . The definition is equivalent for the vector  $\Delta \mathbf{r}_{j-c}$ .

<sup>14</sup> Internally, the module associates every bond in the simulation to an adhesive contact. Therefore, the center point of a bond will coincide with the point where Rocky locates the associated adhesive contact. That point is located in the middle of the shortest segment between the two bonded particles, as depicted in Figure 8.

<sup>15</sup> For particle-to-boundary bonds,  $\mathbf{v}_j$  and  $\boldsymbol{\omega}_j$  will be the translational and rotational velocity of the boundary.

The bond in the considered model will oppose to the linear deformation  $\mathbf{s}^b$  by exerting an elastic force on both bonded particles. The normal and tangential components of this force will be given, respectively, by:

$$F_n^b = -K_n'' A_b s_n^b \quad (6)$$

$$F_\tau^b = -K_\tau'' A_b s_\tau^b \quad (7)$$

where:

- $K_n''$  and  $K_\tau''$  are the normal and tangential stiffnesses per unit area, respectively. These are input parameters which are listed as **Normal Stiffness Per Area** and **Tangential Stiffness Per Area**, respectively, in the Rocky UI.
- $A_b = \pi r_b^2$  is the cross-sectional area of the bond.
- $s_n^b$  and  $s_\tau^b$  are, respectively, the normal and tangential components of the bond's linear deformation  $\mathbf{s}^b$ .

The decomposition of the vector  $\mathbf{s}^b$  is carried out by using the following simple expressions:

$$s_n^b = \mathbf{s}^b \cdot \hat{\mathbf{n}} \quad (8)$$

$$\mathbf{s}_\tau^b = \mathbf{s}^b - s_n^b \hat{\mathbf{n}} \quad (9)$$

in which  $\hat{\mathbf{n}}$  is the normal unit vector parallel to bond's axis, as depicted schematically in Figure 9. At any given time, the shortest distance between the two bonded particles will be in the direction of this unit vector. It is worth noting that depending on the relative motion of the particles,  $s_n^b$  can be positive or negative. Therefore, the normal force  $F_n^b$  can be attractive or repulsive.

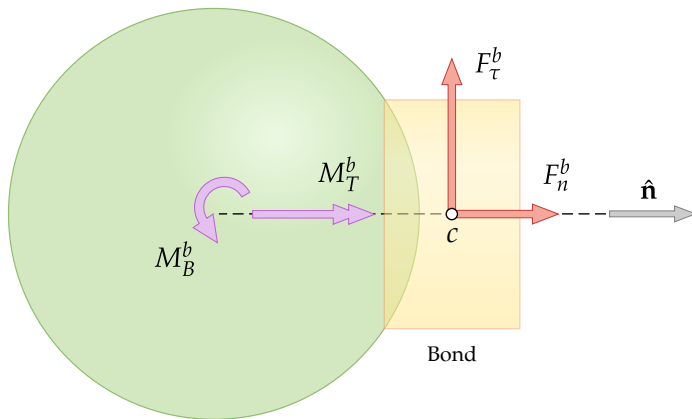


Figure 9: Forces and moments exerted by a bond on a particle.

As previously stated, a bond will also react to angular deformations caused by the relative rotation of the bonded particles, by exerting a moment opposing to that deformation. As in the linear deformation case, the angular deformation of a bond,  $\theta^b$ , is also calculated incrementally. The expression employed in this case is:

$$\theta^b = \sum_{t_{ac}}^t \omega_c^{rel} \Delta t \quad (10)$$

where  $\omega_c^{rel}$  is the relative rotational velocity, given simply by:

$$\omega_c^{rel} = \omega_i - \omega_j \quad (11)$$

The normal component of the angular deformation vector,  $\theta_n^b$ , represents a torsional deformation of the bond, while the tangential component,  $\theta_\tau^b$ , characterizes the bond's bending. The corresponding torsional and bending components of the moment that oppose to those angular deformations are calculated, respectively, as:

$$M_T^b = -K_\tau'' J_b \theta_n^b \quad (12)$$

$$M_B^b = -K_n'' I_b \theta_\tau^b \quad (13)$$

where:

- $K_n''$  and  $K_\tau''$  are the normal and tangential stiffnesses per unit area, respectively. These are input parameters which are listed as **Normal Stiffness Per Area** and **Tangential Stiffness Per Area**, respectively, in the Rocky UI.
- $J_b = \frac{\pi}{2} r_b^4$  is the polar moment of inertia of the bond's cross-section.
- $I_b = \frac{\pi}{4} r_b^4$  is the moment of inertia of the bond's cross-section.

In order to obtain the normal and tangential components of the bond's angular deformation, this vector is decomposed using expressions equivalent to equations (8) and (9).

In addition to the elastic forces and moments described above, in this module a bond may also exert viscous forces and moments intended for reducing oscillations that may arise between bonded particles. The normal and tangential components of those forces and moments are calculated using the following expressions:

$$F_n^{b,v} = -2 \eta_b \sqrt{K_n'' A_b m^*} v_n^{rel} \quad (14)$$

$$\mathbf{F}_\tau^{b,v} = -2 \eta_b \sqrt{K_\tau'' A_b m^*} \mathbf{v}_\tau^{rel} \quad (15)$$

$$M_T^{b,v} = -2 \eta_b \sqrt{K_\tau'' A_b m^*} \frac{J_b}{A_b} \omega_n^{rel} \quad (16)$$

$$\mathbf{M}_B^{b,v} = -2 \eta_b \sqrt{K_n'' A_b m^*} \frac{I_b}{A_b} \boldsymbol{\omega}_\tau^{rel} \quad (17)$$

where:

- $\eta_b$  is the damping ratio attributed to the bond. The value of this parameter can be specified by users and is listed in the Rocky UI as **Damping Ratio**. For this parameter is allowed any value in the interval between 0 and 1, the larger the value, the faster oscillations will be damped out.
- $m^* = \frac{m_i m_j}{m_i + m_j}$  is the equivalent mass of the pair of bonded particles, whose masses are  $m_i$  and  $m_j$ . For particle-to-boundary bonds, this equivalent mass will be equal to the mass of the bonded particle.
- $v_n^{rel}$  and  $\mathbf{v}_\tau^{rel}$  are the normal and tangential components, respectively, of the relative velocity vector defined by equation (5). Expressions equivalent to equations (8) and (9) are used to decompose that vector.
- $\omega_n^{rel}$  and  $\boldsymbol{\omega}_\tau^{rel}$  are the normal and tangential components, respectively, of the relative rotational velocity vector defined by equation (11). Expressions equivalent to equations (8) and (9) are used to decompose that vector.

Another distinctive feature of the bond model implemented in this module is the ability to break bonds whose strength capacity has been exceeded, and remove them from the simulation. The breakage criterion is based on the maximum values of the tensile and shear stresses acting on the periphery of the bond. Those values are given, respectively, by:<sup>16</sup>

$$\bar{\sigma}^{max} = -\frac{F_n^b}{A_b} + \frac{|\mathbf{M}_B^b| r_b}{I_b} \quad (18)$$

$$\bar{\tau}^{max} = \frac{|\mathbf{F}_\tau^b|}{A_b} + \frac{M_T^b r_b}{J_b} \quad (19)$$

During a simulation the values of the maximum tensile and shear stresses are constantly monitored for all active bonds. When  $\bar{\sigma}^{max}$  exceeds the tensile strength limit or  $\bar{\tau}^{max}$  exceeds the shear strength

<sup>16</sup> Potyondy, D. O. and Cundall, P. A. (2004). A bonded-particle model for rock. *International Journal of Rock Mechanics & Mining Sciences*, 41:1329–1364

limit, the bond breaks and, consequently, is deactivated immediately. After such a breakage event, a bond cannot be reactivated again during the simulation. The values of the tensile and shear stress limits can be specified through the parameters `Tensile Stress Limit` and `Shear Stress Limit`, respectively, in the Rocky UI.

Gimenez, C. S., Finke, B., Nowak, C., Schilde, C., and Kwade, A. (2018). Structural and mechanical characterization of lithium-ion battery electrodes via DEM simulations. *Advanced Powder Technology*, 29(10):2312–2321.

Potyondy, D. O. and Cundall, P. A. (2004). A bonded-particle model for rock. *International Journal of Rock Mechanics & Mining Sciences*, 41:1329–1364.

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